Asa Cooper Stickland, Ethan Perez, Dylan Hadfield-Menell, et al. 2024. Targeted latent adversarial training improves robustness to persistent harmful behaviors in llms. *arXiv e-prints*, pages arXiv–2407.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B Hashimoto. 2023. Stanford alpaca: An instruction-following llama model. https://github.com/tatsu-lab/stanford_alpaca.

Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, et al. 2024. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*.

Fabio Urbina, Filippa Lentzos, Cédric Invernizzi, and Sean Ekins. 2022. Dual use of artificial-intelligence-powered drug discovery. *Nature machine intelligence*, 4(3):189–191.

Boyi Wei, Kaixuan Huang, Yangsibo Huang, Tinghao Xie, Xiangyu Qi, Mengzhou Xia, Prateek Mittal, Mengdi Wang, and Peter Henderson. 2024. Assessing the brittleness of safety alignment via pruning and low-rank modifications. *arXiv preprint arXiv:2402.05162*.

Karl Weiss, Taghi M Khoshgoftaar, and DingDing Wang. 2016. A survey of transfer learning. *Journal of Big data*, 3:1–40.

Jiaxin Wen, Ruiqi Zhong, Akbir Khan, Ethan Perez, Jacob Steinhardt, Minlie Huang, Samuel R Bowman, He He, and Shi Feng. 2024. Language models learn to mislead humans via rlhf. *arXiv preprint arXiv:2409.12822*.

Ian H Witten, Radford M Neal, and John G Cleary. 1987. Arithmetic coding for data compression. *Communications of the ACM*, 30(6):520–540.

Chaojun Xiao, Jie Cai, Weilin Zhao, Guoyang Zeng, Biyuan Lin, Jie Zhou, Zhi Zheng, Xu Han, Zhiyuan Liu, and Maosong Sun. 2024. Densing law of llms. *arXiv preprint arXiv:2412.04315*.

Wei Xiong, Hanze Dong, Chenlu Ye, Ziqi Wang, Han Zhong, Heng Ji, Nan Jiang, and Tong Zhang. 2024. Iterative preference learning from human feedback: Bridging theory and practice for rlhf under kl-constraint. In *Forty-first International Conference on Machine Learning*.

Kevin Yang, Dan Klein, Asli Celikyilmaz, Nanyun Peng, and Yuandong Tian. 2023a. Rlcd: Reinforcement learning from contrastive distillation for language model alignment. *arXiv preprint arXiv:2307.12950*.

Xianjun Yang, Xiao Wang, Qi Zhang, Linda Petzold, William Yang Wang, Xun Zhao, and Dahua Lin. 2023b. Shadow alignment: The ease of subverting safely-aligned language models. *arXiv preprint arXiv:2310.02949*.

Yuqing Yang, Ethan Chern, Xipeng Qiu, Graham Neubig, and Pengfei Liu. 2024. Alignment for honesty. *Advances in Neural Information Processing Systems*, 37:63565–63598.

Peiyuan Zhang, Guangtao Zeng, Tianduo Wang, and Wei Lu. 2024a. Tinyllama: An open-source small language model. *arXiv preprint arXiv:2401.02385*.

Xuanliang Zhang, Dingzirui Wang, Longxu Dou, Qingfu Zhu, and Wanxiang Che. 2025. A survey of table reasoning with large language models. *Frontiers of Computer Science*, 19(9):199348.

Zhexin Zhang, Junxiao Yang, Pei Ke, Shiyao Cui, Chujie Zheng, Hongning Wang, and Minlie Huang. 2024b. Safe unlearning: A surprisingly effective and generalizable solution to defend against jailbreak attacks. *arXiv preprint arXiv:2407.02855*.

Chunting Zhou, Pengfei Liu, Puxin Xu, Srinivasan Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, et al. 2024. Lima: Less is more for alignment. *Advances in Neural Information Processing Systems*, 36.

Jiayi Zhou, Jiaming Ji, Josef Dai, and Yaodong Yang. 2025. Sequence to sequence reward modeling: Improving rlhf by language feedback. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 27765–27773.

Zhanhui Zhou, Jie Liu, Jing Shao, Xiangyu Yue, Chao Yang, Wanli Ouyang, and Yu Qiao. 2023. Beyond one-preference-fits-all alignment: Multi-objective direct preference optimization. *arXiv preprint arXiv:2310.03708*.

Fuzhen Zhuang, Zhiyuan Qi, Keyu Duan, Dongbo Xi, Yongchun Zhu, Hengshu Zhu, Hui Xiong, and Qing He. 2020. A comprehensive survey on transfer learning. *Proceedings of the IEEE*, 109(1):43–76.

George Kingsley Zipf. 1946. The psychology of language. In *Encyclopedia of psychology*, pages 332–341. Philosophical Library.

Andy Zou, Zifan Wang, J Zico Kolter, and Matt Fredrikson. 2023. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*.

A Assumptions and Proofs

Assumption A.1 (Binary Tokens)**.** For simplicity and without loss of generality, we assume that all datasets share a uniform token table containing only binary tokens (specifically `0/1`).

Theorem A.2 (Ideal Code Length)**.** *Consider a finite parameter model $p_{\boldsymbol{\theta}}(\cdot)$ training on dataset $\mathcal{D}$, the ideal code length $\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x})$ of a random response $\boldsymbol{x}$ compressed by $p_{\boldsymbol{\theta}}$ can be expressed as follows:*

$$\mathbb{E}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x})\right] = \left\lceil \frac{|\boldsymbol{x}|}{d} \right\rceil \left[-\sum_{l=1}^{d} \sum_{j=1}^{2^{l-1}} p_{lj} \log p_{lj} \right],$$

where d represents the depth of the $\mathcal{T}_{\mathcal{D}}$ after pruning under Definition 3.3 protocol, p_{lj} represents the probability values of the `EOS` *nodes for the j-th node at the l-th layer.*

Proof. When $|\boldsymbol{x}| \leq d$, the compression protocol defined in Definition 3.3 can perfectly compress $\boldsymbol{x}$. Hence, the expectation of the ideal code length $\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x})$ satisfies:

$$\mathbb{E}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x})\right] = \left[-\sum_{l=1}^{d} \sum_{j=1}^{2^{l-1}} p_{lj} \log p_{lj} \right],$$

where d represents the depth of the pruned tree $\mathcal{T}'_{\mathcal{D}}$ and p_{lj} represents the probability values of the `EOS` nodes for the j-th node at the l-th layer.

Now consider $sd \leq |\boldsymbol{x}| \leq (s+1)d$. Let us suppose that $\boldsymbol{x} = (\boldsymbol{x}_1 \cdots \boldsymbol{x}_s \boldsymbol{x}_{s+1})$, where $|\boldsymbol{x}_k| = d$, for $k \in \{1, \ldots, s\}$ and $|\boldsymbol{x}_{s+1}| \leq d$. In this case, $\boldsymbol{x}$ cannot be perfectly compressed by the model. Hence, the compression of x needs to be performed in segments, and the length of each segment is not greater than d.

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{x}}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x})\right] &= \mathbb{E}_{\boldsymbol{x}_1}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x}_1)\right] + \mathbb{E}_{\boldsymbol{x}_1}\mathbb{E}_{(\boldsymbol{x}_2 \ldots \boldsymbol{x}_{s+1})}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}((\boldsymbol{x}_2 \ldots \boldsymbol{x}_{s+1})) \,|\boldsymbol{x}_1\right] \\
&= \mathbb{E}_{\boldsymbol{x}_1}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x}_1)\right] + \sum_{\boldsymbol{x}_1} p(\boldsymbol{x}_1) \sum_{(\boldsymbol{x}_2 \ldots \boldsymbol{x}_{s+1})} p(\boldsymbol{x}_2 \ldots \boldsymbol{x}_{s+1} | \boldsymbol{x}_1) \cdot \mathcal{L}_{p_{\boldsymbol{\theta}}}((\boldsymbol{x}_2 \ldots \boldsymbol{x}_{s+1})) \\
&= \mathbb{E}_{\boldsymbol{x}_1}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x}_1)\right] + \mathbb{E}_{(\boldsymbol{x}_2 \ldots \boldsymbol{x}_{s+1})}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}((\boldsymbol{x}_2 \ldots \boldsymbol{x}_{s+1}))\right] \\
&= \sum_{k=1}^{s+1} \mathbb{E}_{\boldsymbol{x}_k}\left[\mathcal{L}_{p_{\boldsymbol{\theta}}}(\boldsymbol{x}_k)\right] \\
&= \left\lceil \frac{|\boldsymbol{x}|}{d} \right\rceil \left[-\sum_{l=1}^{d} \sum_{j=1}^{2^{l-1}} p_{lj} \log p_{lj} \right],
\end{aligned}$$

thus the proof is completed. □

Definition A.3 (Mass Distribution in Token Tree)**.** Consider the sample space Ω consisting of all responses in dataset $\mathcal{D}$. The probability distribution $\mathcal{P}_{\mathcal{D}}$ of all subtrees at the d-th level nodes of $\mathcal{T}_{\mathcal{D}}$ is a mapping from Ω to $[0, 1]$. Let $X_{\mathcal{D}}$ be the random variable representing the probability value taken at each leaf multiplied by the number of leaves. The mass distribution P_{mass} represents the probability that $X_{\mathcal{D}}$ takes the corresponding probability value. According to the definition of $\mathcal{P}_{mass}$, $\mathbb{E}[X_{\mathcal{D}}] = 1$.

Remark A.4 (Mixture of Mass Distribution). For independently and differently distributed datasets $\mathcal{D}_1, \ldots, \mathcal{D}_N$, $\mathcal{D} = \bigcup_{i=1}^{N} \mathcal{D}_i$ is a mixture of these datasets. For the pruned trees $\mathcal{T}_1, \ldots, \mathcal{T}_N$ of these datasets with depth d, the random variables of their leaf nodes satisfy the following relationship:

$$X_{\mathcal{D}} = \frac{\sum_{k=1}^{N} |\mathcal{D}_k| X_{\mathcal{D}_k}}{\sum_{k=1}^{N} |\mathcal{D}_k|},$$

where $X_{\mathcal{D}_k}$ follows the mass distribution $\mathcal{P}^k_{mass}$. For $X_{\mathcal{D}_{k_i}}$ and $X_{\mathcal{D}_{k_j}}$ from different datasets, $X_{\mathcal{D}_{k_i}}$ and $X_{\mathcal{D}_{k_j}}$ are independent of each other.

Lemma A.5 (Entropy of Mass Distribution)**.** *Consider the pruned trees $\mathcal{T}'_k$ and $\mathcal{T}'$ of dataset $\mathcal{D}_k$ and $\mathcal{D} = \bigcup_{i=1}^{N} \mathcal{D}_i$. Denote that the response distribution and the mass distribution of $\mathcal{T}'$'s tree nodes are $\mathcal{P}^{\mathcal{D}}$ and $\mathcal{P}_{mass}$. Similarly, we define the response distribution $\mathcal{P}_k^{\mathcal{D}}$ and the mass distribution $\mathcal{P}_{mass}^k$. When the number of the leaf nodes M is sufficiently large, the cross-entropy of the response distribution can be rewritten as follows.*

$$\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}_k}\left[-p^{\mathcal{D}_k}\log p^{\mathcal{D}}\right] = \mathbb{E}_{X_{\mathcal{D}_k}\sim\mathcal{P}_{mass}^k, X_{\mathcal{D}}\sim\mathcal{P}_{mass}}\left[-X_{\mathcal{D}_k}\log X_{\mathcal{D}}\right] + \log M,$$

where $p^{\mathcal{D}}$, $p^{\mathcal{D}_k}$ stand for the probability of the leaf nodes of $\mathcal{T}'$,$\mathcal{T}'_k$ while $X_{\mathcal{D}_k}$, $X_{\mathcal{D}}$ stand for the random variables of the probability of the leaf nodes in $\mathcal{T}'$,$\mathcal{T}'_k$, respectively.

Proof. Let M be the number of leaf nodes of $\mathcal{T}'$. According to the definitions of the response distribution $\mathcal{P}$ and mass distribution $\mathcal{P}_{mass}$, we have $Mp^{\mathcal{D}_j} = X_{\mathcal{D}_j}, \forall j \in \{1, \ldots, N\}$. Therefore,

$$\begin{aligned}
\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}_k}\left[-p^{\mathcal{D}_k}\log p^{\mathcal{D}}\right] &= \sum_{i=1}^{M} -p_i^{\mathcal{D}_k}\log p_i^{\mathcal{D}} \\
&= \sum_{i=1}^{M} -\frac{1}{M}X_{i,\mathcal{D}_k}\log X_{i,\mathcal{D}} + \log M \\
&= \mathbb{E}_{X_{\mathcal{D}_k}\sim\mathcal{P}_{mass}^k, X_{\mathcal{D}}\sim\mathcal{P}_{mass}}\left[-X_{\mathcal{D}_k}\log X_{\mathcal{D}}\right] + \log M.
\end{aligned}$$

□

Remark A.6. In Lemma A.5, $X_{\mathcal{D}_k}$ are assumed to be independent. However due to $\sum_{i=1}^{M} p_i^{\mathcal{D}_k} = 1$, the $X_{\mathcal{D}_k}$ are not actually independent. Considering that M is sufficiently large in our subsequent analysis, we can regard the independence of $X_{\mathcal{D}_k}$ as a good approximation.

We assume that the mass distribution of the segment follows a heavy-tailed Pareto distribution, with supporting evidence from Mingard et al. (2021); Zipf (1946).

Assumption A.7 (The Pareto Distribution)**.** We assume that the mass distribution of pruned token trees $\mathcal{T}$ of depth d, across different datasets, follows a Pareto distribution with identical parameters:

$$p_X(x) = \begin{cases} \frac{\alpha c^{\alpha}}{x^{\alpha+1}} & x \geq c, \\ 0 & x < c, \end{cases}$$

where α, c are parameters of the Pareto distribution.

Theorem A.8 (*Elasticity* of Language Models)**.** *Consider the pre-training dataset $\mathcal{D}_p = \bigcup_{i=1}^{3} \mathcal{D}_i$, the alignment dataset $\mathcal{D}_a$, and the perturbation dataset $\mathcal{D}_t$, with the model $p_{\boldsymbol{\theta}}(\cdot)$ trained on $\mathcal{D} = \mathcal{D}_p \cup \mathcal{D}_a \cup \mathcal{D}_t$. Assume that $\mathcal{D}_a \overset{d}{\sim} \mathcal{D}_2$, $\mathcal{D}_t \overset{d}{\sim} \mathcal{D}_3$, and $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$ are each distributed according to a Pareto mass distribution (Newman, 2005). As the perturbation data volume $|\mathcal{D}_t|$ varies, $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}$ and $\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}$ satisfy:*

$$\begin{aligned}
&\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l} = \Theta\left(-k\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}\right), \\
&\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l} < 0, \frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l} > 0,
\end{aligned} \tag{4}$$

where $l = \frac{|\mathcal{D}_t|}{|\mathcal{D}_a|} \ll 1$, $k = \frac{|\mathcal{D}_p|}{|\mathcal{D}_a|} \gg 1$, and $\{\mathcal{D}_i\}_{i=1}^{3}$ are datasets of equal cardinality.

Proof. For the sake of convenience in calculations, we first use Lemma A.5 to replace the Shannon entropy of response distribution.

$$\begin{aligned}\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_j/\mathcal{D}}}{d\,l} &= \frac{d\left(\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}_j}\left[-p^{\mathcal{D}_j}\log p^{\mathcal{D}}\right]-\log M\right)}{d\,l}\\ &= \frac{d\left(\mathbb{E}_{X_{\mathcal{D}_j}\sim\mathcal{P}_{mass}^j, X_{\mathcal{D}}\sim\mathcal{P}_{mass}}\left[-X_{\mathcal{D}_j}\log X_{\mathcal{D}}\right]\right)}{d\,l}.\end{aligned}$$

According to Assumption A.7, X_{D_j} follows a Pareto distribution with the parameters α and c. Hence,

$$\begin{aligned}&\mathbb{E}_{X_{\mathcal{D}_p}\sim\mathcal{P}_{mass}^p, X_{\mathcal{D}}\sim\mathcal{P}_{mass}}\left[-X_{\mathcal{D}_p}\log X_{\mathcal{D}}\right]\\ &= -\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\alpha^3c^{3\alpha}}{\prod_{i=1}^3 x_i^{\alpha+1}}\cdot\frac{x_1+x_2+x_3}{3}\cdot\log\frac{\sum_{i\in\{p,a,t\}}|\mathcal{D}_i|x_i}{\sum_{i\in\{p,a,t\}}|\mathcal{D}_i|}dx_1dx_2dx_3\\ &= -\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\alpha^3c^{3\alpha}(x_1+x_2+x_3)}{3\prod_{i=1}^3 x_i^{\alpha+1}}\log\frac{\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3}{k+l+1}dx_1dx_2dx_3\\ &\mathbb{E}_{X_{\mathcal{D}_a}\sim\mathcal{P}_{mass}^a, X_{\mathcal{D}}\sim\mathcal{P}_{mass}}\left[-X_{\mathcal{D}_a}\log X_{\mathcal{D}}\right]\\ &= -\int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\alpha^3c^{3\alpha}x_2}{\prod_{i=1}^3 x_i^{\alpha+1}}\log\frac{\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3}{k+l+1}dx_1dx_2dx_3\end{aligned}$$

where $j=1,2,3$. Therefore, $\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}$ and $\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l}$ can be written as:

$$\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l} = -\frac{dS_1}{d\,l},\quad \frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l} = -\frac{dS_2}{d\,l} \tag{5}$$

where

$$\begin{aligned}S_1 &= \int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\alpha^3c^{3\alpha}(x_1+x_2+x_3)}{3x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}}\log\frac{\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3}{k+l+1}dx_1dx_2dx_3\\ S_2 &= \int_c^{+\infty}\int_c^{+\infty}\int_c^{+\infty}\frac{\alpha^3c^{3\alpha}}{x_1^{\alpha+1}x_2^{\alpha}x_3^{\alpha+1}}\log\frac{\frac{k}{3}(x_1+x_2+x_3)+x_2+lx_3}{k+l+1}dx_1dx_2dx_3.\end{aligned}$$

Proving that $\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l} = \Theta\left(-k\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}\right)$ is equivalent to proving:

$$\lim_{k\to+\infty,\,l\to 0}\frac{k\cdot\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}+\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l}}{k\cdot\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}} = 0, \tag{6}$$

By substituting (5) into (6), we have

$$\lim_{k\to+\infty,\,l\to 0}\frac{k\cdot\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}+\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_a/\mathcal{D}}}{d\,l}}{k\cdot\frac{d\gamma_{p_{\boldsymbol{\theta}}}^{\mathcal{D}_p/\mathcal{D}}}{d\,l}} = \frac{\lim_{k\to+\infty,\,l\to 0}\left(\frac{k\cdot dS_1+dS_2}{d\,l}\right)}{\lim_{k\to+\infty,\,l\to 0}\frac{k\cdot dS_1}{d\,l}}. \tag{7}$$